

Joint Training of Latent-Space Energy-Based Models and β -VAEs for Anime Face Generation

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KNOWLEDGE
TECHNOLOGY

Outline

- 1 Motivation and Idea
- 2 Method
- 3 Results
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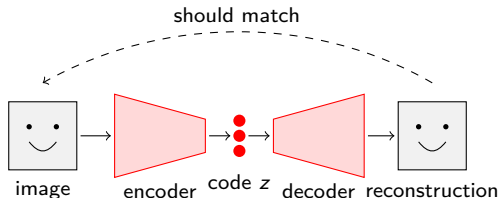
The Problem: Expressive vs. Tractable

- Generative models must be both **expressive** and **tractable** to train.
- **VAEs**: a structured latent space, but often blurry samples.
- **EBMs**: flexible and multi-modal, but an **intractable normalizer** and slow **pixel-space** MCMC sampling (Markov Chain Monte Carlo).

Guiding question

Can we get the *structure* of a VAE and the *precision* of an EBM, affordably?

Variational Autoencoders

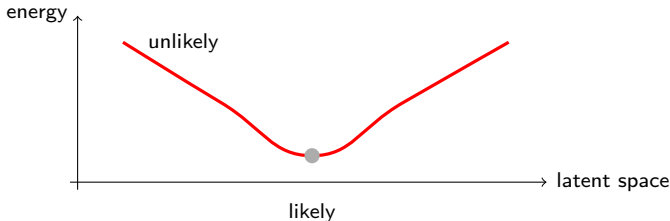


Squeeze an image into a small code z , then rebuild it.

- Training balances two pulls: **reconstruct** the input faithfully vs. keep the latent space **smooth and organized**, so new codes decode into real-looking faces.
- β -**VAE**: turn up the knob β to favour a **cleaner, more disentangled** latent space, each direction controls one intuitive feature.

Energy-Based Models

- An EBM is just a **scoring function**: hand it a latent code, it returns one number, the **energy**.
- **Low energy = looks like a real face**, high energy = doesn't. That's the whole idea.



The model is free to carve valleys wherever real faces happen to live.

- **Generate** a face by rolling **downhill with a little noise** into a valley.
- In a compact 256-D latent space (not pixels), this search stays **fast**.

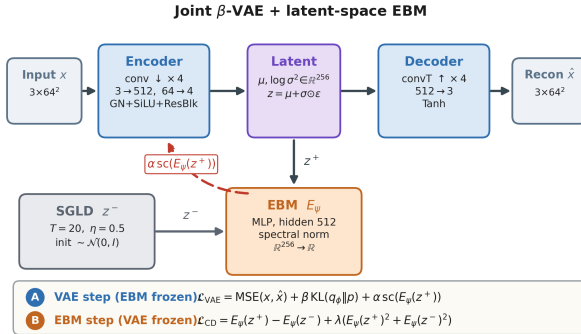
Our Idea: Joint, Bidirectional Coupling

- Prior latent-EBM work is **sequential**: the EBM is added *after* the VAE (a post-hoc prior).
- We train the β -VAE and a latent-space EBM **jointly**.

A two-way street in latent space

- **EBM** $\rightarrow$ **VAE**: an energy penalty shapes the posterior *during* training.
- **VAE** $\rightarrow$ **EBM**: a structured latent manifold makes sampling cheap.

Architecture



- Encoder / decoder: convolutional, with GroupNorm, SiLU, and residual blocks; latent $z \in \mathbb{R}^{256}$.
- EBM: a small MLP with **spectral norm**, only **2.6% of parameters**, discriminative not generative.

Training Step A — VAE Update (EBM frozen)

- **Pretraining first:** VAE alone for 30 epochs (β annealed $0 \rightarrow 1$), so the EBM has a well-formed latent space to score.
- Joint training then **alternates two steps** every iteration: Step A updates the **VAE** while the **EBM** is frozen.

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathcal{L}_{\text{MSE}}}_{\text{stay faithful}} + \beta \underbrace{D_{\text{KL}}(q_{\phi}(z|x) \parallel p(z))}_{\text{stay organized}} + \alpha \underbrace{\text{sc}(E_{\psi}(z^{+}))}_{\text{look real to the EBM}}$$

- The **energy penalty** pulls encoded codes z^{+} into the EBM's low-energy valleys; $\text{sc}(x) = 10 \tanh(x/10)$ keeps it from exploding.
- α is ramped $0 \rightarrow 1$ over the first third: the EBM only starts steering the encoder once it has learned something worth following.

Training Step B — EBM Update (VAE frozen)

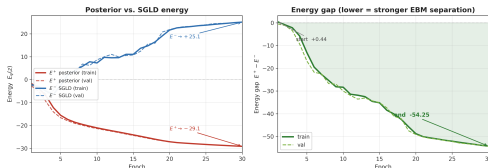
- The VAE is frozen, the EBM learns to tell real codes from fake ones.

$$\mathcal{L}_{\text{CD}} = \underbrace{E_{\psi}(z^{+})}_{\text{push down}} - \underbrace{E_{\psi}(z^{-})}_{\text{push up}} + \lambda \underbrace{(E_{\psi}(z^{+})^2 + E_{\psi}(z^{-})^2)}_{\text{keep energies finite}}$$

- **Positive** z^{+} : encoder outputs, what a *real* face looks like in latent space.
 - **Negative** z^{-} : 20 SGLD steps ($\eta = 0.5$) starting from $\mathcal{N}(0, I)$, the model's own guesses, rolled downhill with noise.
 - Contrastive divergence digs **valleys** under z^{+} and raises **hills** under z^{-} .
- ⇒ a steadily **growing energy gap** between the two.

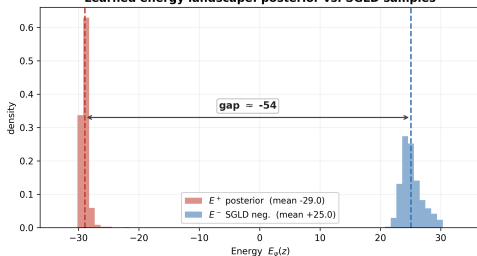
Result: A Growing Energy Gap

EBM energy dynamics over joint training



Energy gap over training

Learned energy landscape: posterior vs. SGLD samples



- Gap grows
 $+0.44 \rightarrow -54.25$; train $\approx$ val (no overfitting).

- **No posterior collapse:**
reconstruction ($708 \rightarrow 691$)
and KL ($151 \rightarrow 142$) stay stable.

Generated Samples



(a) EBM, epoch 5



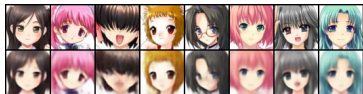
(b) EBM, epoch 30



(c) Gaussian baseline

- EBM-guided (a,b) via 20-step SGLD; baseline (c) decodes $z \sim \mathcal{N}(0, I)$.
- Progression epoch 5 $\rightarrow$ 30 tracks the widening energy gap.

Reconstructions & Latent Interpolation



Reconstructions, epoch 30

- Fidelity retained after joint training; smooth semantic transitions.
- ⇒ a **well-structured latent manifold**.

Latent-space interpolation between real faces (left & right columns are reconstructed endpoints)



Interpolation between real faces

Where We Sit

Method	EBM space	Training	MCMC target	VAE coupling
Pang et al. [2020]	Latent	Sequential	Prior $p(z)$	Inference only
VAEBM [2021]	Pixel	Sequential	Data $p(x)$	None
Nie et al. [2021]	Latent	Sequential	Latent (ODE)	None
Ours	Latent	Joint	Posterior $q_\phi(z x)$	Bidirectional

- The only method with **joint** training and **bidirectional** coupling, the EBM targets the *posterior*.

Conclusion

What we did:

- Jointly trained a β -VAE + latent-space EBM with **bidirectional** coupling.
- A two-phase schedule (β -anneal + α -ramp) kept training stable.

What we found:

- Energy gap opened up ($+0.44 \rightarrow -54.25$): the EBM learned to tell real codes from fake ones.
- No posterior collapse, and reconstruction quality stayed intact.

Takeaway: coupling the two models lets each one help the other, the VAE gives the EBM a clean latent space to work in, and the EBM sharpens what the VAE generates.

The End

Thank you for your attention.
Any questions?

Selected references:

- Kingma, Welling. Auto-Encoding Variational Bayes. *ICLR*, 2014.
- Du, Mordatch. Implicit Generation & Modeling with EBMs. *NeurIPS*, 2019.
- Pang et al. Learning Latent Space Energy-Based Prior Model. *NeurIPS*, 2020.
- Xiao et al. VAEBM. *ICLR*, 2021. Nie et al. *NeurIPS*, 2021.

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